

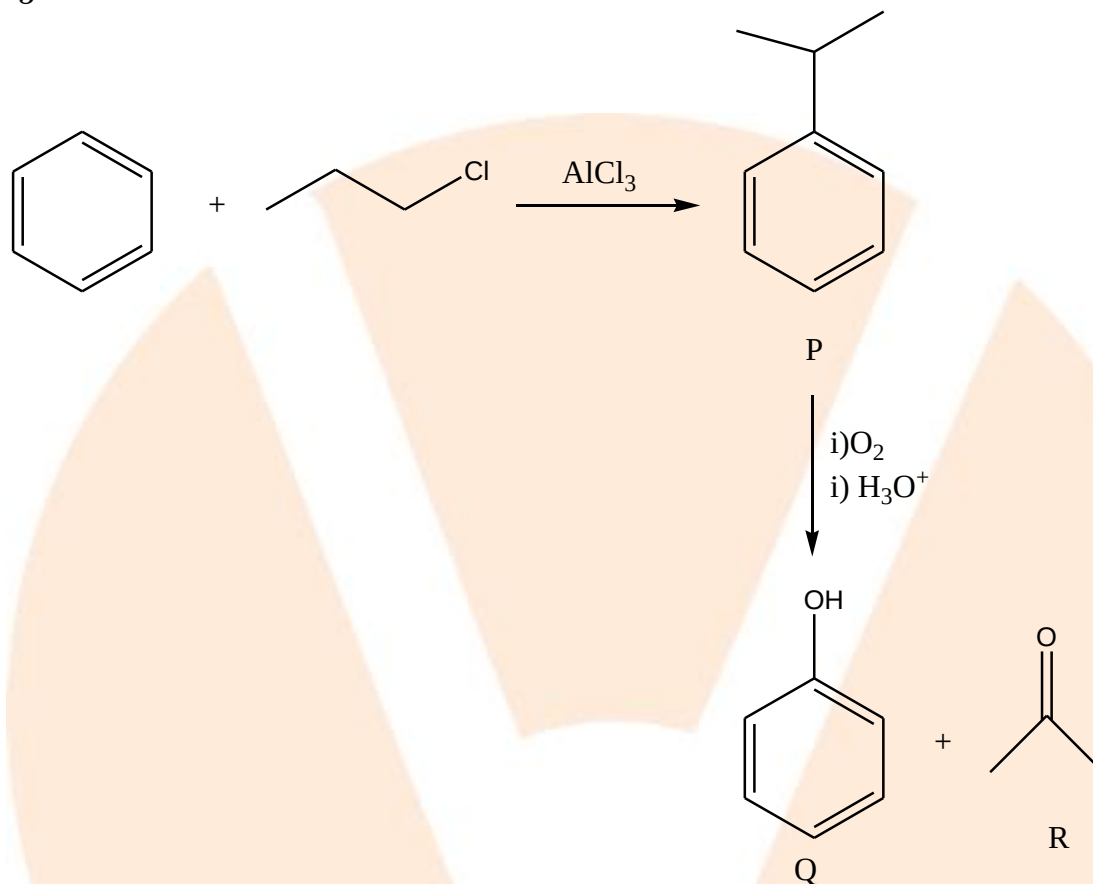
# NEET 2018

Set NN

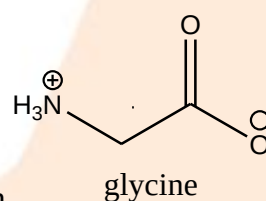
## Solution – Chemistry

NEET -2018 solutions

136.



The above reaction is a cumene reaction . Hence the correct option is 1  
137.



Amino acids can form zwitter ion , i.e. dipolar ion

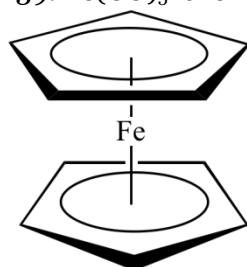
Hence the correct option is 1

138.

$[\text{Ni}(\text{CO})_4]$  is diamagnetic and assumes tetrahedral structure.

Carbonyl is a strong ligand and involves force pairing. Hence correct option is 4

139.  $\text{Fe}(\text{CO})_5$  is ferrocene and is dinuclear complex



Hence the correct option is 1

140.

$\text{Co}^{3+}$  contains 4 unpaired electrons,  $\mu = \sqrt{24}$  B.M.

$\text{Cr}^{3+}$  contains 3 unpaired electrons,  $\mu = \sqrt{15}$  B.M.

$\text{Fe}^{3+}$  contains 5 unpaired electrons,  $\mu = \sqrt{35}$  B.M.

$\text{Ni}^{2+}$  contains 2 unpaired electrons,  $\mu = \sqrt{8}$  B.M.

Therefore the correct option is 2

141.

$[\text{CoCl}_2(\text{en})_2]$  shows geometrical isomerism

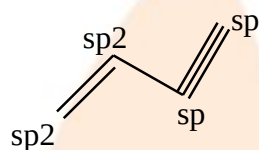
Hence the correct option is 2

142.

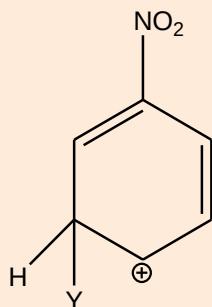
$\text{MnO}_4^{2-}$ , the oxidation state of Mn is +6. It will have one unpaired electron. Hence it will exhibit paramagnetic behavior and also involves d-d transition.

Hence the correct option is 1

143.



Hence the correct option is 4



144.

Hence the correct option is 2.

145.

The correct order of  $-I$  effect is  $\text{NH}_2 < \text{OR} < \text{F}$

Hence the correct option is 2

146.  $\text{BaSO}_4 \rightleftharpoons \text{Ba}^{2+} + \text{SO}_4^{2-}$

$K_{sp} = [\text{Ba}^{2+}] [\text{SO}_4^{2-}]$

$K_{sp} = S^2$

Hence the correct option is 2

147. The rate of liquification is directly proportion to the Vanderwaals constant value.

$\text{NH}_3$  having the highest value should easily undergo liquification.

The correct option is 2.

148. The correct option is 1

149. The coagulation power depends on both magnitude and sign of the charge of the ion.

Hence the correct option is 3

150.

Fluorine can never have positive oxidation state.

Hence the correct option is 3

151

Mg can be used to reduce alumina according to Ellingham diagram

Hence the correct option is 3

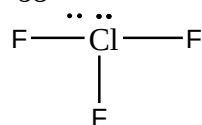
152.

Al is exception in the atomic radii of 13 group

$B < Ga < Al < In < Tl$ .

Hence the correct option is 1

153.



The central Cl atom has 2 lone pair of electrons in  $ClF_3$  molecule.

Hence the correct option is 4

154.

The oxidation state of N is +5, +2, 0, -3 in  $HNO_3$ , NO,  $N_2$ ,  $NH_4Cl$

Therefore the correct option is 2

155.

The maximum coordination number of B is 4. It cannot form  $MF_6^{3-}$

Hence the correct option is 3

156.

A is ethyl alcohol which on reaction with Na forms sodium ethoxide(B).

Ethylalcohol also react with  $PCl_5$  to form chloroethane(C)

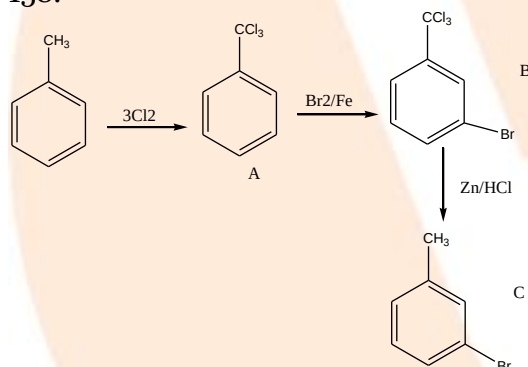
B& C react together to form diethylether. This is Williamson synthesis.

Hence the correct option is 1

157.

Methane undergoes substitution reaction to form ethyl bromide which further undergoes wurtz reaction to form ethane. Hence the correct option is 1

158.



Hence the correct option is 2.

159.

$N_2O_5$  is not a pollutant which can be introduced into atmosphere due to human activity.

The correct option is 2

160.

161.

Metal hydrides are usually ionic in nature.

Smaller size of cation makes it less ionic. So the correct order is

$BeH_2 < CaH_2 < BaH_2$

Hence the correct option is 2

162..

correct option is 1

163. The maximum number of water molecules are present in 18 ml of water.

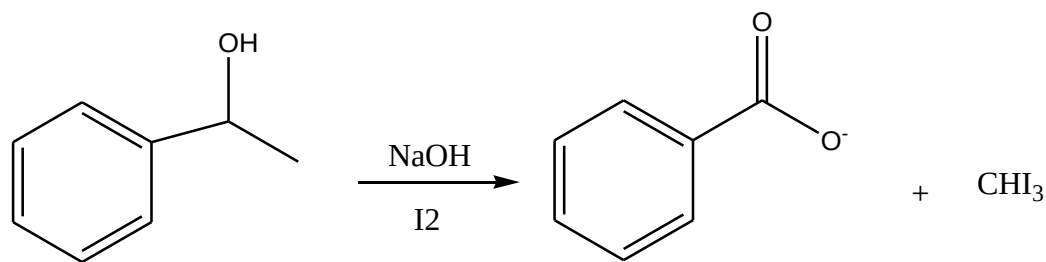
Number of water molecules =  $n \times N_A$

Hence the correct option is 2.

164. Carboxylic acids have higher boiling points than aldehydes and ketones as carboxylic acids can involve intermolecular Hydrogen bonding.

Hence the correct option is 1

165.



A

Hence the correct option is 3  
166.

In Riemer Tiemann Reaction the electrophile is dichlorocarbene.

Hence the correct option is 1

167. Hence the correct option is 3.

168.

The half life period for a zero order reaction is directly proportional to the concn of the reactant. Thus if the concentration of the reactant is doubled the half life period also will be doubled.

Hence the correct option is 4

169.

The correction factor  $a$  to the ideal gas equation corresponds to force of attraction between the gas molecules.

Hence the correct option is 1

170.

Exothermic reactions are best carried out at low temperature and high pressure so that the eqb is shifted towards forward direction.

Hence the correct option is 2

171.

KMnO<sub>4</sub> acts as oxidizing agent in acidic medium and its  $n$  factor is 5. Oxalate ion  $n$  factor is 2.

Hence the correct option is 4

172.

173.

In strong acidic medium aniline converts to anilinium ion and it is a meta director. Therefore the correct option is 1

174. BeO is amphoteric rest all other oxides are basic in nature.

Hence the correct option is 4

175.

Amylose is less branched chain of  $\alpha$  1-4 glycosidic linkage whereas amylopectin is heavily branched and contains  $\alpha$  1-4 glycosidic linkage and  $\beta$  1-6 glycosidic linkage.

Hence the correct option is 3

176.

The correct option is 3.

177.

The electronic configuration written for Nitrogen goes against Hunds rule.

Hence the correct option is 3.

178.

CN<sup>-</sup> has highest bond order 3.

Hence the correct option is 4.

179. X has 5 valence electrons hence it can form X<sup>3-</sup>

The correct option is Mg<sub>3</sub>X<sub>2</sub>.

Hence the correct option is 1

180.

$$\frac{d_1}{d_2} = \frac{z_1}{z_2} \left( \frac{a_2}{a_1} \right)^3$$

The correct option is 3.

